Practical 1

* **Aim:-** Set up Python using a virtual environment, create basic variables, and demonstrate different data types (int, float, string, list, tuple, dict, set).

* **Theory:-** Python is high-level, interpreted programming language that is widely used for software development, data science, web development and automation. Before writing Python programs, it is considered best practice to create a virtual environment. A virtual environment is an isolated workspace that contains a separate Python interpreter and installed libraries. It helps manage project dependencies independently, preventing conflicts between packages used in different projects. In Python, a virtual environment can be created using the command python -m venv myenv and activated before running the program.

A variable is a named memory location used to store a value. Python is dynamically typed, which means the type of a variable is automatically determined based on the value assigned to it. The built-in type() is used to identify the data type of a variable.

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* The following are the datatypes demonstrated in this program:-

① **Integer**:- An integer is used to store whole numbers without decimal points:-
Ex:- age = 20

② **Float**:- A float stores numbers with decimal values.
Ex:- height = 5.9

③ **String**:- String stores textual data enclosed in single or double quotes.
Ex:- name = "Rishabh"

④ **Tuple**:- A tuple is similar to a list but is immutable, meaning its value cannot be changed after creation
Ex:- coordinates = (10,20)

⑤ **List**:- A list is ordered and mutable collection of elements. It can store multiple values of different data types and also allows modification. Ex:- fruits = ["apple", "banana"]

⑥ **Dictionary**:- A dictionary stores data in key-value pairs. It is useful for representing structured information such as student details or records.

⑦ **Sets**:- A set is an unordered collection of unique elements. Duplicate values are automatically removed. It is useful when only distinct values are required.

PAGE NO.
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* Conclusion:- In this practical, a python virtual environment was created and variables of different data types were used. The program demonstrated integers, floats, strings, lists, tuples, dictionaries, and sets and displayed their types using type() function.

Practical 2

* **Aim:-** Write a script that performs arithmetic operations, string manipulations (concatenation, slicing), and type conversions.

* **Theory:-** Python provides built-in support for performing arithmetic calculations, manipulations strings, and converting data from one type to another. These features are essential for developing programs that process numbers and text efficiently.

*Arithmetic Operations:-

Arithmetic operations are used to perform mathematical calculations on numeric values. Python supports operators suchas:-

(i) + for addition
(ii) - for subtraction
(iii) * for multiplication
(iv) / for division
(v) % for modulus

* String Manipulation:-

Strings are used to store text data. Python allows various operations on string including:-

(i) Concatenation:- Joining two or more stings
(ii) Slicing:- Extracting a part of string using index position.

* Type Conversion:-
Type conversion is the process of changing one data type into ~~som~~ another. Python includes built-in functions such as :-
(i) int() to convert to integer
(ii) float() to convert to float.
(iii) str() to convert to string

* Conclusion:- In this practical, arithmetic operations, string manipulation, and type conversion were performed using Python. The program demonstrate mathematical calculations, string concatenation and slicing and conversion between string, integer, and float datatype.